# BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

RESEARCH / PROTOTYPE SYSTEM DISCLAIMER: This system is a research/prototype fingerprint-based blood-group classification system. It is not clinically validated and must not be used as a substitute for standard serological blood-group testing or for medical/transfusion decisions. All predictions and advisory indicators are provided solely for evaluation purposes.

## 1. EXAMINATION & SAMPLE AUDIT DETAILS

| | | | |
|---|---|---|---|
| **Report ID:** | BG-000602 | **Timestamp:** | 2026-09-08 21:42:47 |
| **Input Source:** | Upload | **Evaluation Status:** | **LOW_CONFIDENCE** |
| **Model Name:** | Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%) | **Confidence:** | **48.20%** |
| **Advisory Status:** | **NEEDS CONFIRMATION** (τ=70%) | **Advisory Reason:** | **LOW_CONFIDENCE_AND_RH_UNCERTAINTY** |

## 2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP

# Predicted Blood Group:

# AB+

Confidence Score: **48.20%**

Advisory Threshold: **70%**

Advisory Screening: **NEEDS CONFIRMATION**

Derived ABO Group: **AB** (48.3% prob) | Rh Factor: **+**

**RESEARCH ADVISORY:** Advisory threshold passed (≥ 70.0%). Note: Confidence OK does not constitute medical verification. Standard serological testing required for medical/transfusion use.

**Rh FACTOR UNCERTAINTY ADVISORY:** Rh factor (±) determination is uncertain (moderate 8-class confidence (48.2%)). ABO group appears to be 'AB' (48.3% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

## 3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

| A+ | A- | AB+ | AB- | B+ | B- | O+ | O- |
|---|---|---|---|---|---|---|---|
| 0.1% | 42.8% | 48.2% | 0.1% | 5.0% | 0.0% | 3.6% | 0.2% |